

Contact

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Education

🎓 Bachelor's -- Computer Science
🏛 Rochester Institute of Technology
📅 December, 2015 - May, 2017

Skills

Programming Languages

Scala, Clojure(script),
Java, SQL, Rust, Python,
Javascript

Infrastructure

PostgreSQL, Snowflake,
Airflow, MySQL, Kafka, Linux

Devops

Docker, Terraform, GCP, AWS,
CircleCI, Bamboo

Monitoring

Grafana, Prometheus,
BugSplat, Slack Integration

Theory

Functional Programming,
Object Oriented
Programming,
Type Theory,
Language & Framework
Design,
Typed Functional
Programming

Luke Cook

Senior Backend Software Engineer



Datasembly



Remote

Senior Software Engineer (2022-2024)
Software Engineer (2020-2022)

📅 December 2020 - July 2024

Responsibilities

- Develop and maintain automated data collection systems, written in Scala, for grocery-related datasets
- Maintain Airflow DAGs written in Python
- Reverse engineer web APIs and integrate them into a unified data pipeline
- Design and implement evasion techniques to ensure high data acquisition success rates
- Collaborate with Product and Customer teams to transform raw data into actionable insights

Projects

- Led development of a Playwright-based browser automation framework, simplifying integration logic and supporting advanced features such as JavaScript rendering, API encryption handling, and dynamic login workflows
- Initiated effort to define and standardize the 'Out-of-Stock' detection logic, collaborating with cross-functional teams to align on data parsing patterns. Improved understanding, and clarity into data meaning.



Hyperfiddle



Remote

Software Engineer

📅 March 2020 - September 2020

Responsibilities

- Built full-stack features in a Clojure/ClojureScript + Datomic application framework
- Implemented CI/CD pipeline using CircleCI, AWS ECR/ECS, and Terraform
- Built configurable sample data generation library to support product demos and internal testing workflows

Projects

- Designed an extensible algebra system for Datomic statements
 - Enabled automatic simplification of negating statements (e.g., $1 + -1 = 0$)
 - Developed frontend logic to emit statements that self-simplify on change
 - Proved Datomic statements formed an Algebraic Group structure



Cornell University

Ithaca New York

Application Programmer II

June 2019 - June 2020

Responsibilities

- Improved genomic CSV digester application
 - Simplified instruction file format, reducing redundancy (V2)
 - Rewrote digester to be more generic and powerful, developed declarative Aspect Files (v32)
 - Created a file translator to convert instruction files between V1, V2, and V3 in Clojure(script)

Projects

- Developed framework for automated end-to-end testing
 - Created a domain-specific scripting language for test structure: Arrange/Act, Assert, Cleanup
 - Clojure for backend, Java for business logic, and custom syntax for test specifics
 - Ensured test independence and idempotency in production environments



Intuit

Mountain View, CA

Software Engineer Co-Op

Jan 2018 - July 2018

Responsibilities

- Contributed to an agile team developing microservices for account creation workflows
- Implemented a document upload endpoint for identity verification during onboarding
- Increased test reliability and confidence by expanding unit test coverage across critical components

Projects

eldritch

Algebraic Data Types & Pattern Matching
<https://www.github.com/invocatis/eldritch>

motif

Recursive Pattern Matching in Clojure
<https://www.github.com/invocatis/motif>

ernie

Testing Framework for Java written in Clojure
<https://www.github.com/invocatis/ernie>

Scala 3 Algebraic Type Exploration

Exploration of algebraic concepts defined in Scala 3
<https://github.com/Invocatis/scala3-algebraic-type-exploration>

CSV Digester

CSV ingestion and manipulation library
<https://bitbucket.org/gobiiproject/gobii.masticator/src/develop/>